Dietary patterns and risk of oesophageal cancers: a population-based case-control study.

Epidemiological studies investigating the association between dietary intake and oesophageal cancer have mostly focused on nutrients and food groups instead of dietary patterns. We conducted a population-based case-control study, which included 365 oesophageal adenocarcinoma (OAC), 426 oesophagogastric junction adenocarcinoma (OGJAC) and 303 oesophageal squamous cell carcinoma (OSCC) cases, with frequency matched on age, sex and geographical location to 1580 controls. Data on demographic, lifestyle and dietary factors were collected using self-administered questionnaires. We used principal component analysis to derive three dietary patterns: 'meat and fat', 'pasta and pizza' and 'fruit and vegetable', and unconditional logistic regression models to estimate risks of OAC, OGJAC and OSCC associated with quartiles (Q) of dietary pattern scores. A high score on the meat-and-fat pattern was associated with increased risk of all three cancers: multivariable-adjusted OR 2·12 (95 % CI 1·30, 3·46) for OAC; 1·88 (95% CI 1·21, 2·94) for OGJAC; 2·84 (95% CI 1·67, 4·83) for OSCC (P-trend <0·01 for all three cancers). A high score on the pasta-and-pizza pattern was inversely associated with OSCC risk (OR 0·58, 95 % CI 0·36, 0·96, P for trend=0·009); and a high score on the fruit-and-vegetable pattern was associated with a borderline significant decreased risk of OGJAC (OR for Q4 v. Q1 0·66, 95% CI 0·42, 1·04, P=0·07) and significantly decreased risk of OSCC (OR 0·41, 95% CI 0·24, 0·70, P for trend=0·002). High-fat dairy foods appeared to play a dominant role in the association between the meat-and-fat pattern and risk of OAC and OGJAC. Further investigation in prospective studies is needed to confirm these findings.